



IILM  
UNIVERSITY

SUMMER INTERNSHIP PRESENTATION

# SmartSplit

Expense Management & Payment Settlement Platform

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**Organization:** BitePay

**Duration:** 7 June 2026 – 7 August 2026

**Domain:** AI, Data Analytics & Software Development

# Internship Overview



## Organization

BitePay — Tech that Serves



## Role

AI, Data Analytics & Software Dev. Intern



## Duration

2 Months (7 Jun – 7 Aug 2026)



## Domain

Fintech — Payments & Data Analytics

## Key Areas of Exposure

- Full-Stack Web Application Development
- REST API Design (Node.js + Express.js)
- Database Design with MongoDB & Mongoose
- JWT Authentication & Secure Password Handling
- Payment Gateway Integration (Razorpay)
- Real-World Fintech Product Workflow

# About BitePay



*"Tech that Serves"*

MSME Registered · 2025



BitePay is a fintech company building digital payment and financial-utility products.



Works across AI/Data Analytics and software engineering to power its payment-related software.



Focuses on practical, real-world product building — interns contribute directly to live features.

**My Role:** As an intern in AI, Data Analytics & Software Development, I contributed to building a fintech-style expense and payment-settlement product from the ground up.

# Objectives of the Internship



Build a complete full-stack application end-to-end, not isolated components



Understand secure authentication using JWT and password hashing



Integrate a real third-party payment gateway (Razorpay) into a live workflow



Gain hands-on experience designing and consuming REST APIs



Learn schema design and data modeling with MongoDB



Develop practical exposure to a professional fintech product environment

# Project: SmartSplit

*A full-stack expense-splitting and payment-settlement application*

## What it does

SmartSplit simplifies managing shared expenses. Users create shared expenses, split amounts equally or by custom shares among participants, track balances on a dashboard, and settle outstanding amounts online.

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Payments are handled through Razorpay Checkout, and every payment is verified on the backend rather than trusted from the frontend alone — closing a common security gap in payment flows.



### User Accounts

Registration & secure login



### Expense Splitting

Equal or custom shares among participants



### Dashboard

Track expenses & outstanding balances



### Payments

Settle shares via Razorpay Checkout

# Technologies & Tools Used



## Frontend

React.js, JavaScript, HTML, CSS



## Backend

Node.js, Express.js (REST APIs)



## Database

MongoDB with Mongoose ODM



## Auth & Security

JWT authentication, bcrypt password hashing



## Payments

Razorpay Payment Gateway & Checkout

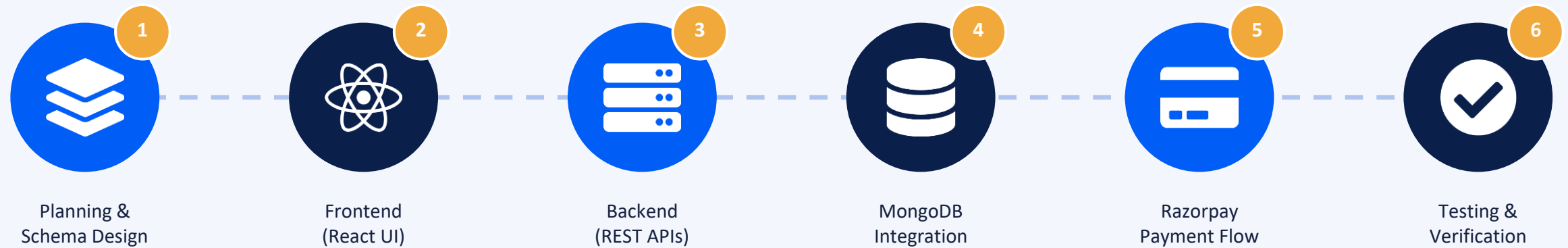


## Dev Tools

VS Code, npm, Nodemon, Git/GitHub, dotenv,  
CORS

# Development Workflow

*How the frontend, backend, database and payment workflow came together*



*Integrated the frontend, backend APIs, database and payment workflow into a single working application — rather than building components in isolation.*

# Challenges & Key Learnings



## Challenges

- MongoDB Atlas DNS/SRV resolution and TLS connection issues when connecting the Node.js backend to the Atlas cluster
- Understanding how connection strings, DNS resolution, IP whitelisting and Node.js config affect a live backend
- Integrating the Razorpay flow — creating orders, launching Checkout from React, and securely verifying payments on the backend



## Key Learnings

- How to build a full-stack app end-to-end — connecting React, Express, and MongoDB instead of isolated pieces
- Practical experience designing REST APIs and implementing JWT-based authentication
- Designing MongoDB schemas for a real application
- How to integrate and secure a real online payment workflow



# Internship Outcomes

*After completing the two-month internship, I was able to:*



Deliver a working full-stack SmartSplit prototype — React dashboard + Node/Express backend + MongoDB



Enable users to register, log in, and manage shared expenses end to end



Support equal and custom expense splitting among participants



Let users view outstanding balances and initiate real payments via Razorpay



Implement backend payment verification, not just frontend trust



Gain practical, integrated experience across auth, database design and fintech payment workflows



# Conclusion

Successfully completed a 2-month Summer Internship in AI, Data Analytics & Software Development at BitePay, from 7 June 2026 to 7 August 2026.

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Built SmartSplit, a full-stack expense-splitting and payment-settlement application connecting React, Express, MongoDB and Razorpay — gaining hands-on experience in authentication, database design, REST APIs and real-world fintech payment integration.

*“This internship strengthened my ability to connect theoretical knowledge with real-world, end-to-end product development — and deepened my interest in building practical, full-stack fintech solutions.”*